MAP-A

A Landmark-Anchored Timetable-Graph Model of Public Transport: Verified Computation of Accessibility, Its Inequality, and the Fare Matrix from Open Data, with Agent-Reported Hospital Bed Capacity for Ambulance and Patient Routing



MATHEMATICS AND COMPUTATIONAL SCIENCES TEAM (MCS-T)

Brimbuela, Peter John A.
De Asis, Dave Justin Ll.
Shin, Yeaseong

Legazpi City Science High School

Checklist for Adult Sponsor (1)

This form is required for ALL projects.
This form must be completed by the adult sponsor BEFORE experimentation.

To be completed by the Adult Sponsor in collaboration with the student researcher(s):

Student's Name(s): Yeasong Shin, Peter John A. Brimbuela, Dave De Asis

Project Title: An Open Timetable-Graph Model of Informal Public Transport: Measuring Accessibility and Inequality in Albay Province

1. ☑ I have reviewed the ISEF Rules and Guidelines, including the science fair ethics statement.
2. ☑ I have reviewed the student's completed Student Checklist (1A) and Research Plan/Project Addendum.
3. ☑ I have worked with the student and we have discussed the possible risks involved in the project.
4. ☐ The project involves one or more of the following and requires prior approval by an SRC, IRB, IACUC or IBC:
   - ☐ Humans
   - ☐ Vertebrate Animals
   - ☐ Potentially Hazardous Biological Agents
     - ☐ Microorganisms ☐ rDNA ☐ Tissues
5. ☑ Items to be completed for **ALL PROJECTS**
   - ☑ Adult Sponsor Checklist (1)
   - ☑ Student Checklist (1A)
   - ☑ Research Plan/Project Addendum
   - ☑ Approval Form (1B)
   - ☑ Student Support Disclosure Form (2A)
   - ☐ Regulated Research Institutional/Industrial Setting Form (2C) (when applicable; after completed experiment)
   - ☐ Continuation/Research Progression Form (7) (when applicable)

Additional forms required if the project includes the use of one or more of the following (check all that apply):

☐ **Humans,** including student designed inventions/prototypes. (Requires prior approval by an Institutional Review Board (IRB); see full text of the rules.)
- ☐ Human Participants Form (4) or appropriate Institutional IRB documentation
- ☐ Sample of Informed Consent Form (when applicable and/or required by the IRB)
- ☐ Qualified Scientist Form (2B) (when applicable and/or required by the IRB)

☐ **Vertebrate Animals** (Requires prior approval, see full text of the rules.)
- ☐ Vertebrate Animal Form (5A)-for projects conducted in a school/home/field research site (SRC prior approval required
- ☐ Vertebrate Animal Form (5B)-for projects conducted at a Regulated Research Institution. (Institutional Animal Care and Use Committee (IACUC) approval required prior experimentation.)
- ☐ Qualified Scientist Form (2B) (Required for all vertebrate animal projects at a regulated research site or when applicable)

☐ **Potentially Hazardous Biological Agents** (Requires prior approval by SRC, IACUC or IBC, see full text of the rules.)
- ☐ Potentially Hazardous Biological Agents Risk Assessment Form (6A)
- ☐ Human and Vertebrate Animal Tissue Form (6B)-to be completed in addition to Form 6A when project involves the use of fresh or frozen tissue, primary cell cultures, blood, blood products and body fluids.
- ☐ Qualified Scientist Form (2B) (when applicable)
- ☐ The following are exempt from prior review but require a Risk Assessment Form 3: projects involving protists, archae and similar microorganisms, for projects using manure for composting, fuel production or other non-culturing experiments, projects using color change coliform water test kits, microbial fuel cells, and projects involving decomposing vertebrate organisms.

☐ **Hazardous Chemicals, Activities and Devices**
- ☐ Risk Assessment Form (3)
- ☐ Qualified Scientist Form (2B) (as applicable, see full text of the rules)

☑ **Other**
- ☑ Risk Assessment Form (3)

☑ **I attest to the information checked above and that I have read and agree to abide by the science fair ethics statement.**

DICK ANTHONY M. CALTEJA		07/03/26
Adult Sponsor's Printed Name	Signature	Date of Review (mm/dd/yy)
09171075074	dickanthony.calteja@deped.gov.ph	
Phone	Email	

Student Checklist (1A)

This form is required for ALL projects.
This form must be completed by the student(s) BEFORE experimentation.

1. a. Student/Team Leader: Yeaseong Shin Grade: 12

   Email: shin.yeaseong@gmail.com Phone: 09998371113

   b. Team Member: Peter John Brimbuela c. Team Member: Dave De Asis

2. Title of Project: An Open Timetable-Graph of Informal Public Transport: Measuring Accessability and Inequality in Albay Province

3. School: Legazpi City Science High School School Phone: 09922300719
   (if multiple schools, list of the team leader or list all schools).

   School Address: Bitano, Legazpi City

4. Adult Sponsor: Dick Anthony M. Calleja Phone/Email: 0917075074

5. Where will you conduct your experimentation? (check all that apply)

   ☐ Research Institution ☑ School ☐ Field ☑ Home ☐ Other: _____

6. List the name and address of all non-home and non-school work site(s), whether you worked there virtually or on-site:

   Name _____ _____

   Address _____ _____

   _____ _____

   Phone/email _____ _____

7. Does this project need SRC/IRB/IACUC or other pre-approval? ☐ Yes ☒ No Tentative start date: 01/03/26

8. This year's experimentation/data collection:

   07/06/26 (Actual Start Date: (mm/dd/yy)) 08/14/26 (End Date: (mm/dd/yy))

9. Is this a continuation/progression from a previous year? ☐ Yes ☒ No

   a. If yes, attach the previous year's ☐ Abstract **and** ☐ Research Plan/Project Addendum

   b. Explain how this project is new and different from previous years on

   ☐ Continuation/Research Progression Form (7); include forms for all previous years

10. Source of Data:

    ☑ self-collected ☐ mentor collected ☑ publicly available ☐ Other

Provide a complete Research Plan/Project Addendum with this form.

Research Plan/Project Addendum

A Research Plan/Project Addendum is required for ALL projects. The plan must be written BEFORE experimentation.

1. The Research Plan is to be written prior to experimentation.
2. If changes are made during the research before competing at an affiliated fair, such changes must be added as an addendum to the original plan. Be aware that some changes may require obtaining or returning to the IRB or SRC for appropriate review and approvals before conducting that portion of the project. If there are no changes, an addendum is not required.
3. The Research Plan/Project Addendum should include the following:
   a. **RATIONALE**: Include a brief synopsis of the background that supports your research question/goal.
   b. **RESEARCH QUESTION(S), HYPOTHESIS(ES), ENGINEERING GOAL(S), EXPECTED OUTCOMES**: Include a brief synopsis of the background that supports your research question/goal.
   c. Describe the following in detail (include any required subject-specific additions to your Research Plan (listed below):
      - **List of materials:**
      - **Procedures/Methodology:** Detail all procedures and steps of the experimental design including methods for data collection, and when applicable, the source of data used. Be sure to differentiate steps to be taken by you and the steps that are to be taken by any supervising adult(s).
      - **Risk and Safety:** Identify any potential risks and describe safety precautions to be taken.
      - **Data Analysis:** Describe the procedures you will use to analyze the data/results.
   d. **ACKNOWLEDGEMENTS:** List all individuals that supported your project and the contributions that they made. If AI was used, document this support with a prompt log or an explanation of the elements of the project that involved AI support.
   e. **BIBLIOGRAPHY:** List major references (e.g. science journal articles, books, internet sites) from your literature review. Ensure that all references are properly listed and verified (The use of AI for citations is discouraged; it is known to hallucinate and produce false citations which becomes an ethical issue if you cite sources that are false or that you did not actually reference.)

Items 1-4 below are subject-specific guidelines for additional items to be included in your Research Plan/Project Addendum as applicable.

1. **Human participants research:**
   a. **Participants:** Describe age range, gender, racial/ethnic composition of participants. Identify vulnerable populations (minors, pregnant women, prisoners, mentally disabled or economically disadvantaged).
   b. **Recruitment:** Where will you find your participants? How will they be invited to participate?
   c. **Methods:** What will participants be asked to do? Will you use any surveys, questionnaires or tests? If yes and not your own, how did you obtain? Did it require permissions? If so, explain. What is the frequency and length of time involved for each subject?
   d. **Risk Assessment:** What are the risks or potential discomforts (physical, psychological, time involved, social, legal, etc.) to participants? How will you minimize risks? List any benefits to society or participants.
   e. **Protection of Privacy:** Will identifiable information (e.g., names, telephone numbers, birth dates, email addresses) be collected? Will data be confidential/anonymous? If anonymous, describe how the data will be collected. If not anonymous, what procedures are in place for safeguarding confidentiality? Where will data be stored? Who will have access to the data? What will you do with the data after the study?
   f. **Informed Consent Process:** Describe how you will inform participants about the purpose of the study, what they will be asked to do, that their participation is voluntary and they have the right to stop at any time.
2. **Vertebrate animal research:**
   a. Discuss potential ALTERNATIVES to vertebrate animal use and present justification for use of vertebrates.
   b. Explain potential impact or contribution of this research.
   c. Detail all procedures to be used, including methods used to minimize potential discomfort, distress, pain and injury to the animals and detailed chemical concentrations and drug dosages.
   d. Detail animal numbers, species, strain, sex, age, source, etc., include justification of the numbers planned.
   e. Describe housing and oversight of daily care.
   f. Discuss disposition of the animals at the end of the study.
3. **Potentially hazardous biological agents research:**
   a. Give source of the organism and describe BSL assessment process and BSL determination.
   b. Detail safety precautions and discuss methods of disposal.
4. **Hazardous chemicals, activities & devices:**
   a. Describe Risk Assessment process, supervision, safety precautions and specific methods of disposal.
   b. Safety Data Sheets must be reviewed and should be referenced but do not need to be submitted with paperwork unless required by your fair.

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MATHEMATICS AND COMPUTATIONAL SCIENCES TEAM (MCS-T)

Brimbuela, Peter John A.
De Asis, Dave Justin Ll.
Shin, Yeaseong

Legazpi City Science High School

RESEARCH PLAN

Title: A Landmark-Anchored Timetable-Graph Model of Public Transport: Verified Computation of Accessibility, Its Inequality, and the Fare Matrix from Open Data, with Agent-Reported Hospital Bed Capacity for Ambulance and Patient Routing (MAP-A)

Researchers: Brimbuela, Peter John A.; De Asis, Dave Justin Ll.; Shin, Yeaseong

Adviser: Calleja, Dick Anthony M.

Category: Mathematics and Computational Sciences — Team (MCS-T)

Rationale

No published, machine-readable record exists for the routes, stops, headways, or fares of public transport in Albay Province, which runs on jeepneys, tricycles, minibuses, and UV vans. Without a formal network model, nobody can say how many essential places, such as hospitals, schools, public markets, and government offices, can be reached from a given point within a reasonable travel time, or how unevenly that access is spread across the province's 18 local government units (LGUs) and 1,374,768 residents (Philippine Statistics Authority [PSA], 2021).

The gap matters beyond academic interest. The Public Utility Vehicle Modernization Program requires each LGU to prepare a Local Public Transport Route Plan backed by evidence (Department of Transportation [DOTr], 2017; Department of the Interior and Local Government [DILG], 2018), yet no such evidence exists for Albay. Official registers also record only the authorized beds of a hospital, not the free ones, so a patient or ambulance crew cannot tell which nearby hospital can actually admit them, offers the level of care needed, and fits the patient's ability to pay. Earlier projects that opened informal transit data, such as Digital Matatus in Nairobi (Williams et al., 2015), began with field mapping. This study asks how much can be defined, proven, and computed from open

data before any field campaign. An open, verified method for measuring access and its inequality would support route planning, the siting of facilities, emergency routing, and evacuation planning in a province exposed to typhoons and Mayon Volcano, in line with Sustainable Development Goal targets 11.2, 3.8, and 10.2 (United Nations General Assembly [UNGA], 2015).

Research Question or Problem Being Addressed

Given a machine-readable model of the public transport network of Albay's first to third legislative districts, built solely from publicly available sources, how many essential landmarks can be reached from each anchor point within a given door-to-door travel time, and how evenly is that number distributed across anchor points and municipalities?

Specifically, the study seeks to answer the following:

> **Q1.** If there are no official schedules, can departure times be estimated from service headways, and can a time-expanded timetable graph answer earliest-arrival queries to every landmark correctly and efficiently on ordinary hardware?
>
> **Q2.** How does a vehicle-class constraint (width, vertical clearance, gross mass, axle load) change the set of reachable destinations, and can the resulting road networks be ordered so that routing for many vehicle classes stays computationally cheap?
>
> **Q3.** Under the regulated fare structure, is it still computationally tractable to route on fare, arrival time, and number of transfers at once?
>
> **Q4.** Do the chosen inequality measures keep their expected properties when accessibility is computed over a register of landmarks instead of a full population partition, and can the Theil index still separate inequality within and between municipalities?
>
> **Q5 (hospital extension).** Can lightweight software agents collect hospital bed-status reports through standard e-mail protocols while keeping message volume low, keeping reports fresh, and discarding invalid or forged replies?

Q6 (hospital extension). If a patient is matched to hospitals by acuity, required service level, reported bed status, and payment class, does an earliest-arrival search find the nearest suitable facility exactly, and how does the recommendation change as urgency increases?

Goals/Expected Outcomes/Hypotheses

- To construct a time-expanded timetable graph of Albay's public transport from OpenStreetMap roads, public facility registers, and estimated headways, and to publish it as a GTFS feed that passes the open-source GTFS Schedule validator.
- To define landmark-anchored cumulative accessibility at a 30-minute baseline, with an exponential-decay measure as a robustness check, and to summarize its inequality using a population-weighted Gini coefficient, a Theil index decomposed by municipality, Lorenz curves, and the Palma ratio.
- To state and prove twelve mathematical results covering query correctness, computational complexity, vehicle-class ordering, inequality-measure properties, fare monotonicity and subadditivity, exact early stopping in hospital search, monotone hospital suitability with the payment gap as a corollary, and bounds on agent message load and report age.
- To verify every result computationally against a separately written ground-truth implementation on randomized synthetic test cases, and to demonstrate the whole pipeline on a synthetic seven-stop instance (S1) with known correct answers.
- To build a hospital mode that routes ambulances and patients to the nearest hospital that has the required service level, a bed that is not reported full, and a cost class the patient can bear, with the payment filter lifted for emergencies as Republic Act No. 10932 requires.

The following model hypotheses were fixed before the study-area instance was run. They are statements about the open-data model and need no field measurement:

- **M1.** Modeled access to hospital beds has a weighted Gini above 0.30 at the 30-minute baseline (bootstrap interval wholly above 0.30).
- **M2.** Urban-center anchors show greater modeled accessibility than rural anchors.
- **M3.** The between-municipality share of the Theil index is under half of the total.
- **M4.** A larger vehicle class never reaches more of the road network than a smaller one (exact by Proposition 3; any violation flags a faulty road attribute).
- **M5.** The directions of M1 to M3, M6, and M7 hold under every anchor set and every travel budget of 15, 30, and 45 minutes.
- **M6.** For a stable patient (acuity 3) needing free or assisted care, the payment gap is larger at rural anchors than at urban anchors.
- **M7.** For acuity 2, more than half of the anchor weight has a plain-nearest hospital that fails the capability test.
- **M8.** No anchor shows a non-zero payment gap for emergency profiles (exact by Corollary 11; any violation flags a software error).

Materials and Procedures

No laboratory materials, chemicals, or field equipment were used. The study used the following data and tools:

- OpenStreetMap road network extract for the Bicol Region (OpenStreetMap contributors, 2026)
- PSA 2020 Census of Population and Housing and the Philippine Standard Geographic Code
- Department of Health National Health Facility Registry, hospital licensing records, and Malasakit Center directory; PhilHealth list of accredited hospitals; Department of Education Masterlist of Schools; Philippine Ports Authority port listings

- Land Transportation Franchising and Regulatory Board fare adjustment effective September 28, 2026 (Daguno-Bersamina, 2026; Nieto, 2026) and the 20 percent statutory fare discount
- Personal and school computers; a programming environment with standard scientific and graph-algorithm libraries; the open-source GTFS Schedule validator (MobilityData)

1. Road Network

The road network was built from OpenStreetMap as a weighted directed graph. Each road segment carries a length, a free-flow speed, and four limits: usable width, vertical clearance, gross mass, and axle load. Limits that were never recorded were set to infinity and flagged. Route geometries connecting the main market centers and transit hubs were generated computationally under vehicle-class constraints.

2. Landmark Verification and Mapping

Candidate landmarks were taken from the public registers under inclusion criteria fixed before listing began. Anchors are trip starting points (terminals, municipal and city halls, plazas, ports, major boarding junctions); essential destinations are hospitals and health units, public schools, government offices, public markets, and evacuation centers. Duplicates within 50 m were merged, each landmark was snapped to its nearest road node, and any landmark more than 150 m from the road network was flagged and removed rather than moved. Barangay populations from the 2020 census were summed into anchor weights.

3. GTFS Feed Generation

Stops, routes, trips, stop times, calendars, fares, and shapes were compiled into a GTFS feed, together with landmark and hospital extension files. The feed had to pass the GTFS Schedule validator with no blocking errors before use.

4. Graph Construction and Queries

The feed and walking links (4.5 km/h, at most 15 minutes per walk, at least 4 minutes per transfer) were compiled into a time-expanded timetable graph following Pyrga et al. (2008). Earliest-arrival times were computed with a label-setting shortest-path search (Dijkstra, 1959) and averaged over 25 departure times from 06:00 to 08:00.

5. Accessibility and Inequality

Travel times were converted into cumulative landmark accessibility at a 30-minute baseline (Wachs & Kumagai, 1973), with an exponential-decay measure as a robustness check (Hansen, 1959). Scores were summarized by a population-weighted Gini coefficient (Cowell, 2011), a Theil index decomposed by municipality (Theil, 1967; Shorrocks, 1980), Lorenz curves, and the Palma ratio.

6. Hospital Layer Integration

Each hospital and infirmary received a record of service capability level following the Department of Health (2012) classification, emergency-room flag, ownership, PhilHealth accreditation, Malasakit Center presence, cost class, and published official contact address. Patient acuity follows the five-level Emergency Severity Index (Gilboy et al., 2011) and is entered by the user or ambulance crew; the application performs no triage.

7. Live Bed Tracking (Simulated)

Six types of software agents (registry, coordinator, inquiry, reader, validator, and status) were built to send bed-status inquiries by SMTP every 8 hours with at most one reminder, read replies by IMAP, validate them against five rules, and publish free beds with the age of each report. This stage was run only against a simulated mail server with 60 synthetic hospitals over 30 days; no real hospital was contacted.

8. System Prototype

The outputs were integrated into a prototype web application with routing, fare information, vehicle-class restrictions, and a hospital mode. The prototype is still under development and was not tested by external users as part of this study.

9. Mathematical Proofs and Verification

Twelve propositions and corollaries were formally stated and proven. Each was then verified against a separately written ground-truth method (exhaustive enumeration, Floyd–Warshall distances, or a dynamic program in topological order) on randomized synthetic test cases, and demonstrated end to end on the synthetic instance S1, which does not represent real conditions in Albay.

Risk and Safety

This is a computational and mathematical study. No chemicals, hazardous devices, biological agents, or human subjects were involved. No survey or interview was carried out, and no hospital was contacted. All data came from public sources or are explicitly labeled synthetic instances built for verification, including the simulated mail server.

The principal risks are methodological. The risk that synthetic results are mistaken for real-world measurements is reduced by labeling every model output and by never presenting S1 values, including its hospital attributes and bed counts, as findings about Albay or any real facility. The risk that a software shortcut silently produces wrong routes is reduced by verifying every core result against independent reference code before adoption.

For hospital mode, the application does not diagnose or triage; clinical judgment always takes precedence, and every hospital-mode screen carries a one-tap 911 call (Executive Order No. 56, 2018). Before any real hospital receives an inquiry, the protocol and message text must be approved by the Scientific Review Committee, the Provincial Health Office must be informed, and each facility must opt in. Only aggregate bed counts

are requested, no patient information is collected, and facility addresses are handled under the Data Privacy Act of 2012. Standard ergonomic precautions for extended computer use apply. A completed Risk Assessment Form (3) is attached.

Data Analysis

Computed results were checked against independently written reference implementations on randomized synthetic test cases, and agreement rates and error magnitudes were recorded. Earliest-arrival results were compared with exhaustive enumeration of feasible journeys (618 queries over 120 synthetic timetables, and 400 queries on S1). The ambulance router was compared with Floyd–Warshall all-pairs distances and the public-transport hospital router with a dynamic program in topological order (5,200 queries in total). The agent network was run for 30 simulated days with 200 forged replies, logging messages per cycle, opt-out handling, report age, and validation outcomes. Query times were measured on networks of increasing size and fitted on a log-log scale.

Statistical treatment for the open-data model instance was pre-registered. Eight questions use a population-weighted Gini with 95% percentile bootstrap intervals from 2,000 resamples (Efron & Tibshirani, 1993), the Mann–Whitney U test with rank-biserial effect size (Mann & Whitney, 1947; Kerby, 2014), Theil decomposition with a bootstrap interval for the within-group share, Spearman rank correlation (Spearman, 1904), median discrete Fréchet distance between Lorenz curves (Eiter & Mannila, 1994), and headway perturbation of ±25% and ±50%. Decisions are made at $\alpha = 0.05$ with Holm (1979) correction across the family of tests. A conclusion is treated as robust only if its direction holds across every sensitivity sweep. Because these tests run on model outputs, their significance describes the model, not measured road conditions.

Acknowledgements

Generative AI (Claude) was used to assist with explaining and summarizing mathematical concepts from the team's own research in plain language; drafting administrative materials for submission (logbook entries, this research plan, and Risk Assessment language) based on the team's actual project history and decisions; giving feedback on presentation slides and scripts and creating illustrative diagrams for the logbook; and coding the prototype web application. The mathematical framework, propositions and proofs, software implementation, and paper were developed by the student team; the AI tool assisted with organizing, explaining, and drafting supporting materials around that work.

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RESEARCH PLAN ADDENDUM

The following changes were made after the District Science and Technology Fair research plan was submitted.

1. Replacement of the Field Survey

The original plan was a volunteer-led field survey in which volunteers would ride transit routes end to end while recording GPS traces, boarding and alighting times, fares, and vehicle types. Given a team of three researchers and the project timeline, a multi-week, multi-route GPS campaign was judged infeasible, and it would have required human-subjects review not otherwise needed by the computational core. It was replaced by computational network construction from OpenStreetMap under vehicle-class constraints. The change affects only how the road and transit layer is sourced; the mathematical framework, proofs, and verification are unaffected. No additional SRC/IRB approval was required, since the change removed human-subject involvement.

2. Refocus on Landmarks

The unit of analysis changed from barangays to landmarks listed in public registers. Barangay populations are used only as anchor weights, and no barangay receives an accessibility value of its own. The study area covers all three legislative districts (18 LGUs).

3. Addition of the Hospital Extension

The model was extended with hospital records, an e-mail agent network for bed availability, and a hospital mode that recommends the nearest suitable hospital for ambulances and patients, together with four new proven results and their verification. The extension reads the same graph, vehicle classes, and landmark set. The agent network was tested only against a simulated mail server. Live deployment requires prior SRC approval of the protocol and message text, notice to the Provincial Health Office, and opt-in consent from each facility.

4. Updated Fare Parameters

The LTFRB fare increase scheduled for March 19, 2026 was deferred and took effect on September 28, 2026. The fare model now uses the parameters in effect from that date.

The original field-survey protocol is retained as a proposed method for future field validation, should empirical data collection be undertaken as a follow-on project.

Approval Form (1B)

This form is required for ALL projects (one form per student).
This form must be signed by the student researcher and parent BEFORE experimentation.
The SRC will sign as appropriate for pre-approval and/or approval just before competition.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and agree to uphold all aspects of the student researcher ethics statement.

Student researchers are expected to maintain the highest standards of honesty and integrity. Scientific fraud and misconduct are not condoned at any level of research or competition. Such practices include but are not limited to plagiarism, forgery, use or presentation of other researcher's work as one's own, and fabrication of data. Fraudulent projects will fail to qualify for competition in affiliated fairs and ISEF.

Yeaseong Shin		06/11/26
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the Research Plan/Project Addendum. I consent to my child participating in this research.

Youngsoon Shin		06/11/26
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB approval. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Addendum** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Addendum** before the student begins experimentation.

SRC Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution **(not home or high school, etc.)**, was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (2C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval (Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Addendum** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Approval Form (1B)

This form is required for ALL projects (one form per student).
This form must be signed by the student researcher and parent BEFORE experimentation.
The SRC will sign as appropriate for pre-approval and/or approval just before competition.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and agree to uphold all aspects of the student researcher ethics statement.

Student researchers are expected to maintain the highest standards of honesty and integrity. Scientific fraud and misconduct are not condoned at any level of research or competition. Such practices include but are not limited to plagiarism, forgery, use or presentation of other researcher's work as one's own, and fabrication of data. Fraudulent projects will fail to qualify for competition in affiliated fairs and ISEF.

PETER JOHN A. BRIMBUELA	Peter John	06/10/26
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Addendum.** I consent to my child participating in this research.

FRELAN S. BRIMBUELA	Jim	06/10/26
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB approval. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Addendum** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Addendum** before the student begins experimentation.

SRC Chair's Printed Name

Signature

Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution **(not home or high school, etc.)**, was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (2C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature

Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval (Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair

I certify that this project adheres to the approved **Research Plan/Project Addendum** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Approval Form (1B)

This form is required for ALL projects (one form per student).
This form must be signed by the student researcher and parent BEFORE experimentation.
The SRC will sign as appropriate for pre-approval and/or approval just before competition.

1. To Be Completed by Student and Parent

a. Student Acknowledgment:

- I understand the risks and possible dangers to me of the proposed research plan.
- I have read the ISEF Rules and Guidelines and will adhere to all International Rules when conducting this research.
- I have read and agree to uphold all aspects of the student researcher ethics statement.

Student researchers are expected to maintain the highest standards of honesty and integrity. Scientific fraud and misconduct are not condoned at any level of research or competition. Such practices include but are not limited to plagiarism, forgery, use or presentation of other researcher's work as one's own, and fabrication of data. Fraudulent projects will fail to qualify for competition in affiliated fairs and ISEF.

Dave Justin Ll. De Asis		06 / 11 / 26
Student's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

b. Parent/Guardian Approval: I have read and understand the risks and possible dangers involved in the **Research Plan/Project Addendum**. I consent to my child participating in this research.

Dominador Ll. De Asis Jr.		06 / 11 / 26
Parent/Guardian's Printed Name	Signature	Date Acknowledged (mm/dd/yy) (Must be prior to experimentation.)

2. To be completed by the local or affiliated Fair SRC

(Required for projects requiring prior SRC/IRB approval. Sign 2a or 2b as appropriate.)

a. Required for projects that need prior SRC/IRB approval BEFORE experimentation (humans, vertebrates or potentially hazardous biological agents).

The SRC/IRB has carefully studied this project's **Research Plan/ Project Addendum** and all the required forms are included. My signature indicates approval of the **Research Plan/Project Addendum** before the student begins experimentation.

SRC Chair's Printed Name

Signature | Date of Approval (mm/dd/yy) (Must be prior to experimentation.)

OR

b. Required for research conducted at all Regulated Research Institutions with no prior fair SRC/IRB approval.

This project was conducted at a regulated research institution **(not home or high school, etc.)**, was reviewed and approved by the proper institutional board before experimentation and complies with the ISEF Rules. **Attach (2C) and any required institutional approvals (e.g. IACUC, IRB).**

SRC Chair's Printed Name

Signature | Date of Signature (mm/dd/yy) (May be after experimentation)

3. Final ISEF Affiliated Fair SRC Approval (Required for ALL Projects)

SRC Approval After Experimentation and Before Competition at Regional/State/National Fair
I certify that this project adheres to the approved **Research Plan/Project Addendum** and complies with all ISEF Rules.

Regional SRC Chair's Printed Name	Signature	Date of Approval (mm/dd/yy)
State/National SRC Chair's Printed Name *(where applicable)*	Signature	Date of Approval (mm/dd/yy)

Student Support Disclosure Form (2A)

This form is required for ALL projects. This form may be completed before or after experimentation.

It is important to fully disclose any support provided in the completion of a scientific research project. This disclosure provides acknowledgment and verification of the project. Team members should complete one form for the project.

To be completed by the student researcher(s):

Please list the person serving in each role. (Described in the Roles & Responsibilities of Students and Adults section of the Rules.) If an individual serves in multiple roles, list fully once and by name only in other roles. Additional pages can be attached.

1. **Adult Sponsor**			
Name	Organization/School	Email	Phone
Dick Anthony Calleja	Legazpi City Science High School	dick anthony.calleja@deped.gov.ph	09171075074
Relationship to student:	Expertise:	Contribution to project:	Duration of support for this project:
Adviser	PHD in Mathematics Education (Cand		

2. **Direct Supervisor**			
Name	Organization/School	Email	Phone
Dick Anthony Calleja	Legazpi City Science High School	dick anthony.calleja@deped.gov.ph	09171075074
Relationship to student:	Expertise:	Contribution to project:	Duration of support for this project:
Adviser	PHD in Mathematics Education (Cand		

3. **Qualified Scientist**			
Name	Organization/School	Email	Phone
Relationship to student:	Expertise:	Contribution to project:	Duration of support for this project:

4. **Additional Supporting Adults** (graduate students, relatives, external experts, etc.) Use additional pages as necessary.			
Name	Organization/School	Email	Phone
Relationship to student:	Expertise:	Contribution to project:	Duration of support for this project:

Paid Support Disclosure:

1. Did you or your parents pay for any support for this project? ☒ Yes ☐ No
   If yes, explain: (Indicate any fees, tuition and/or payment to any of the individuals above and/or to a program that provided any support for this project including lab space, mentorship, documentation preparation, etc.)
   General expenditure such as papers and printer ink.

AI Usage Disclosure

1. Did your project involve the use of Generative AI? ☒ Yes ☐ No

   If yes, please provide an attachment with Generative AI used in your project and in what elements of your project it was used.

   b. Please provide the AI Prompt Log containing the prompts and responses from the Generative AI or AI Wrapper used. Be sure to remove any personal prompts which contain personal identifying information about yourself or others. If you do not have access to your prompt log, then please provide a summary of the prompts and responses involved in your project.

Attestation by Student Researcher(s):

I certify that I/we have presented the full truth regarding all of the support received in the completion of my/our research project and have not omitted pertinent information nor presented false information.

Finalist Signature: _____ **Date:** 09/04/26

Printed Name(s): Brimbuela, Peter John A.; Brimbuela; Dave De Asis; Yeaseong Shin

Risk Assessment Form (3)

This form is recommended for ALL projects; may be required for projects involving hazards, potentially hazardous biological agents or human participants. This form must be completed BEFORE experimentation.

Student's Name(s) Yeasoong Shin, Peter John A. Brimbuela, Dave D. Ajir

Title of Project An Open Timetable-Graph Model of Informal Public Transport: Measuring Accessibility and Inequality in Albay Province

To be completed by the Student Researcher(s) in collaboration with Direct Supervisor/Qualified Scientist: (All questions must be answered; additional page(s) may be attached.)

1. a) List all hazardous chemicals, activities or devices to be used

   None. Computational/mathematical project only; no chemicals, hazardous devices, or physical hazards used.

   b) List all microorganisms to be used

   Not applicable.

2. Identify and assess the risks and hazards involved in this project.

   Methodological only.

   a. synthetic results being mistaken for real world measurements
   b. unverified code producing slightly incorrect results
   c. standard strain from extended computer use.

3. Describe the safety precautions and procedures that will be used to reduce the risks. If you conducted field work, include permits received and safety plans, as applicable.

   Synthetic results clearly labeled throughout and never presented as real findings; all results are independently verified against a separate ground-truth method before use; standard ergonomics are followed. No field work with humans/animals/regulated sites conducted; no permits applicable.

4. Describe the specific disposal procedures that will be used (when applicable).

   Not applicable. No chemical, biological, or physical waste produced.

5. List the source(s) of safety information.

   Society for Science, International Rules: Guidelines for Science and Engineering Fairs 2026-2027 (societyforscience.org/ISEF).

To be completed and signed by the Direct Supervisor (or Qualified Scientist, when applicable):
I agree with the risk assessment and safety precautions and procedures described above. I certify that I have reviewed the Research Plan/Project Addendum and the International Rules, including the science fair ethics statement and will provide direct supervision.

DICK ANTHONY M. CALLEJA — Direct Supervisor's Printed Name

D.A. Calleja — Signature

07/03/26 — Date of Review (mm/dd/yy)

PhD Math Ed Course - Data Analysis Subject / Quantitative Method, Enhancing Mentoring in Scientific Research
Experience/Training as relates to the student's area of research

TVI / LEGASU
Position/Institution

09171075074 / dickanthony.calleja@deped.gov.ph
Phone or email contact information